

1) Stack and Queue

Stack :

Push (54) $\rightarrow$ (54)

Push (52) $\rightarrow$ [54, 52]

Pop() $\rightarrow$ remove 52

Push (55) $\rightarrow$ [54, 55]

Push (62) $\rightarrow$ [54, 55, 62]

S.Pop() $\rightarrow$ 62

Queue:

enqueue (21) $\rightarrow$ [21]

enqueue (24) $\rightarrow$ [21, 24]

dequeue () $\rightarrow$ removes 21

enqueue (28) $\rightarrow$ [24, 28]

enqueue (32) $\rightarrow$ [24, 28, 32]

q = dequeue () $\rightarrow$ 24

$q = 24$

$\therefore -8 + q = 62 + 24 = 86$

2) Evaluate postfix expression

Expression 10 5 + 60 6 / * 8 -

• $10 + 5 = 15$

• $60 / 6 = 10$

• $15 \times 10 = 150$

$150 - 8 = 142$

Ans: 142

3) postfix 8 2 3 ^ / 2 3 * + 51 -

• $2 3 ^ = 8$

• $8 / 8 = 1$

$2 3 * = 6$

$\therefore \text{Stack} = 1, 6$

4/

$$a + b * c = d^1 2^1 6$$

^ is right associative
* before + and -

postfix $abc*+def^1 2^1 6$

5) Push(10) Push(20) POP(), PUSH(10), PUSH(20), POP(), POP(),
POP(), PUSH(20), POP

Popped values

20
20
10
10
20

[10] $\rightarrow$ 10
[10, 20] $\rightarrow$ 20
[10, 10] $\rightarrow$ 10
[10, 10, 20] $\rightarrow$ 20
[20, 10] $\rightarrow$ 10
[10, 20] $\rightarrow$ 20
[20] $\rightarrow$ 20
[] $\rightarrow$ 10
[20] $\rightarrow$ 20
[20, 10] $\rightarrow$ 10
[10] $\rightarrow$ 10
[] $\rightarrow$ 20

Sequence: 20, 20, 10, 10, 20

4/ char	Stack	O/P
a	Empty	A
+	+	A
b	+	AB
*	+ *	AB
c	+ *	ABC
-	-	ABC +
d	+ -	ABC * + d
^	- ^	ABC * + d
2	- ^	ABC * + d
^	- ^	ABC * + d
6	- ^	ABC * + d
-	-	ABC * + d
empty	empty	ABC * + d